

BSP-Y02 & SIMILAR ARCADE STICKS

DIY Web Arcade Controller Guide for Public & Youth-Center Terminals

Hardware Setup • Browser Gamepad API • Button Roles • Testing

Prepared for Coin Op 4 Charity / Arcade Angels

Making a tablet + arcade stick feel like a reliable free-play station for community spaces

1. Overview & Goals

The BSP-Y02 is a small wireless arcade stick (joystick plus face buttons, Select, Start, and optional Turbo/Macro). On Android tablets and most computers it appears as a standard HID gamepad. A browser can read it with the Gamepad API, so a static website—such as the Coin Op 4 Charity terminal—can be played with a real stick without building a full MAME cabinet.

For public and youth-center use the priorities are the same as any placement: reliability, simple controls, clear exit behavior, and a station that staff can set up without specialized tools. This guide covers physical setup, recommended button roles, the minimal software pattern used on coinop4charity.org, and a checklist before you leave a tablet at a host site.

2. Safety & Practical Limits

- These are consumer electronics, not grounded coin-door arcade panels. Keep drinks away from the stick and tablet; inspect cables regularly.
- The BSP-Y02 phone clamp is narrow (about 48–91 mm). Many tablets are wider—do not force them. Mount the tablet on a stand or shelf and use the stick only as a controller.
- Secure the tablet so it cannot walk away easily. Prefer landscape orientation at a height comfortable for kids and teens.
- If using USB, manage the cable so it is not a trip hazard. Prefer Bluetooth for a clean table when the tablet supports stable HID pairing.

3. Hardware Setup (Beginner)

Charge the stick fully before an event. Power it on and put it in the mode your manual specifies for Android / standard HID (batch manuals differ; common combos involve Home with A or X—verify on your unit).

On the tablet: open Bluetooth settings and pair the device (the name may appear as BSP-Y02, BSP-YOI, or similar). After pairing, reload the game page—many browsers only refresh the gamepad list when a pad connects.

Confirm the system sees the stick with any free web “gamepad tester” page: axes should move with the joystick and buttons should light when pressed. If the browser sees nothing, re-pair in standard HID mode (not a phone key-mapping app mode), try wired USB-C/OTG if available, then reload.

4. How the Browser Sees the Stick

Modern browsers expose `navigator.getGamepads()`. Each pad reports axes (typically axis 0 = left/right, axis 1 = up/down) and buttons with a pressed flag. Your page should poll the gamepad every animation frame while a menu or game is active. No Unity or native app is required for basic free-play web games—plain JavaScript on HTTPS is enough.

Merge stick input with keyboard and on-screen touch controls so the same game works for everyone. One shared state object (for example `gamepadState.left / .fire`) keeps every mini-game consistent.

5. Recommended Button Roles (Public Use)

Arcade sticks invite accidents if one button both shoots and quits. Use clear roles:

Control — Menu / Attract | Inside a game

- Stick / D-pad — Move highlight | Move
- A — Confirm / start game | Primary action (fire, jump, punch)
- B — — | Secondary only (dash, etc.) — never quit
- X / Y / RT — — | Optional extra fire
- Select — Next menu item | Exit to menu
- Start — Confirm | Hold about 0.5 second to exit
- Any stick edge — Wake attract mode | —

Do not map “any button = fire.” If B, Start, and shoulders all shoot, turbo and mash input will feel broken and players will exit by mistake.

6. Deadzone, Hysteresis & Debounce

Cheap sticks and square gates chatter around center. If any tiny axis value counts as left/right, the menu highlight jumps by itself. Use hysteresis: engage a direction only past a higher threshold (about 0.48) and release only below a lower threshold (about 0.32). Debounce menu moves (about 150–180 ms) so one push does not skip three games. Edge-detect “just pressed” for confirm and exit; do not treat a held turbo as repeated menu clicks.

7. Software Pattern Used on the Charity Terminal

Coin Op 4 Charity’s public `index.html` polls the gamepad continuously, keeps a clean `gamepadState`, and applies different logic by context:

- Attract mode: any new stick or button edge closes attract and focuses the game list.
- Menu: stick moves a highlight on game buttons; A or Start launches the focused game.
- In-game: stick and A/B feed the same paths as keyboard and touch; Select exits; Start held ~0.5 s exits; B never exits.
- Optional kiosk URL (`?kiosk=1`) hides contact forms and admin chrome so a dedicated tablet is mostly games + stick hints.

Prefer one site with a kiosk flag over maintaining a second mirror site. Document the live mapping in your repo so future volunteers can adjust thresholds without guessing.

8. Minimal Implementation Outline

1) Create a shared `gamepadState` object (left, right, up, down, fire, jump, dash, a, b, start, select). 2) Each frame, call `navigator.getGamepads()`, read axes and buttons, apply deadzone/hysteresis, set fire from A (and optional X/RT) only. 3) In each game loop, OR gamepad flags with keyboard and touch flags. 4) On the menu, use edge detection for navigation and confirm. 5) On exit, use Select or hold Start—not B. 6) Listen for `gamepadconnected` / `gamepaddisconnected` and show a short “arcade stick online” cue if you want staff feedback.

Full production code lives in the public coinop4charity.org `index.html` input section and in the project GitHub repository. Copy patterns from there rather than inventing a second control scheme.

9. Final Testing Checklist Before Placement

- Stick pairs; page reload shows the pad connected (tester page or on-site badge).
- Attract appears after idle; moving the stick closes attract and focuses a game.
- Highlight moves one step at a time (no skipping from drift or turbo).
- A or Start launches the highlighted game.
- In-game: stick moves; A fires/jumps; B does not quit.
- Select returns to the menu cleanly.
- Hold Start about half a second returns to the menu.

- Disconnect the stick mid-game; touch or keyboard still works.
- Reconnect the stick without rebooting the tablet.
- Tablet is secure, landscape, brightness usable in the room; no trip hazard from cables.
- Staff card on the table lists: A = play, Select = back, hold Start = exit.

10. Common Problems

- **Browser never sees the pad:** Wrong Bluetooth mode or mapping-app mode — re-pair as standard Android HID; try USB; reload the page.
- **Menu focus drifts or jumps:** Deadzone too small — raise engage threshold; add hysteresis and menu debounce.
- **Pressing B quits the game:** B was wired to back and fire — exit only on Select or hold Start.
- **Turbo opens many games:** Menu used held state instead of edges — edge-detect confirms; debounce moves.
- **Works on laptop, not tablet:** Browser or OS blocked gamepad until a gesture — use HTTPS, reload after pair, update the browser.
- **Tablet too wide for the clamp:** Clamp limit ~91 mm — separate mount; stick stays on the table.

11. Notes for Host Sites

Provide a short one-page sheet: how to power the tablet on/off, that play is free, stick button summary (A play, Select back, hold Start exit), and who to contact if the page or stick fails. Bag spare USB cable or charge cable with a label. If the station uses ?kiosk=1, bookmark that URL or the home-screen icon so volunteers do not land on the full public form view by mistake.

These tablet terminals complement restored cabinets; they do not replace locked coin doors or full electromechanical safety practices on vintage machines. Use them where a low-cost, easy-to-move free-play station helps the room feel alive.

12. Document Control

Guide type: DIY controller / web arcade • Related hardware: BSP-Y02 and similar HID arcade sticks • Software reference: coinop4charity.org terminal (Gamepad API, kiosk mode) • Audience: volunteers and beginners setting up youth-center tablet stations • Companion to the restoration PDF series under docs/restoration/.

When you change button thresholds or exit behavior on the live site, update this guide in the same commit so field notes and code stay aligned.